

Name of the Student	N.AKSHAYA
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GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO5

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **N.AKSHAYA(192425129)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.AKSHAYA **Name of the student:** N.AKSHAYA

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS:

2. Stochastic Encoders, Decoders and Contractive Encoders

Prepare a technical presentation explaining Stochastic Encoders and Decoders and Contractive Encoders. Describe their working principles, objectives, robustness characteristics, and differences from standard autoencoders. Demonstrate a suitable real-world application for each approach.

PRESENTATION ANSWER :

Feature	Stochastic Encoder/Decoder	Contractive Encoder
Principle	Introduces randomness during encoding/decoding	Makes representation insensitive to small input changes
Working	Samples latent representations from a probability distribution	Penalizes sensitivity of encoder to input variations
Objective	Learn robust probabilistic representations	Learn stable and robust features
Robustness	Handles uncertainty and input variations	Robust to small perturbations
Standard AE Difference	Uses probabilistic encoding/decoding instead of deterministic mapping	Adds a contraction penalty to the loss
Application	Image generation and uncertain data modeling	Image recognition and robust feature extraction

Working

Stochastic Encoder:

Input → Encoder → Probability Distribution → Random Latent Sample → Decoder → Output.

Contractive Encoder:

Input → Encoder → Stable Latent Representation → Decoder → Reconstruction.

Objective

- **Stochastic:** Learn useful latent representations while modeling uncertainty.
- **Contractive:** Minimize reconstruction error and reduce sensitivity to small input changes.

Real-World Applications

- **Stochastic Encoder/Decoder:** Image generation and probabilistic data modeling.
- **Contractive Encoder:** Robust image recognition and feature extraction.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand